

# Arijit De

Kolkata, India

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## PROFESSIONAL OVERVIEW

Product Engineer with a PhD having 6 years of industry experience spanning Multi-Agent AI systems, Computer Vision and MLOps. Designed and built end to end scalable, robust multi modal RAG pipelines, 3D/2D Image Segmentation and Object Detection products across construction, healthcare, and automotive domains. Proficient in Python, FastAPI, Langchain, CrewAI, PyTorch and cloud deployment (GCP, AWS).

## WORK EXPERIENCE

Product Engineer | mVizn, Singapore | Remote JAN 2024 – PRESENT

- Leading development of a multi-agent platform for construction domain (**CrewAI, Python, FastAPI**) powering Autonomous Project Management & Scheduling (APMS), and intelligent project information retrieval — serving 100 concurrent users / 3000 agent runs per day and increasing business efficiency by 50%.
- Instrumented end-to-end observability, tracing, and performance monitoring with **CrewAI Tracing** and **Ragas**, lifting evaluation scores (faithfulness / answer-relevancy) by 75%.
- Built multi-modal Retrieval-Augmented Generation (**RAG**) pipelines over massive text, image, and CAD datasets (10M+ vectors in **Pinecone**) using hybrid search and reranking, achieving retrieval accuracy (recall@k) of 92% while holding retrieval latency under 600 ms.
- Created a scalable end-to-end machine learning system from training to deployment for 3D semantic segmentation of indoor point clouds extracted from LiDAR scans using Python and PyTorch reducing Building Information Modeling (BIM) creation time by 60%.
- Leveraged **MLflow** for model versioning, model registry, and reproducible training workflows cutting experimentation tracking time by 70%.
- Designed and implemented scalable, robust backend architectures using **FastAPI**, Pydantic, asyncio and multi threaded programming.
- Used Google Cloud Platform (**GCP**) services like Cloud Storage, Cloud Run, IAM, RBAC to deploy and maintain dynamic scalability with 99% uptime.

AI Product Engineer | Synerjix, USA | Remote JUN 2025 – DEC 2025

- Built a Smart Moving Assistant that estimates shifting cost of items from room photos, improving stakeholder business efficiency by 50%, adopted by 3500 users / 10000 estimates processed.
- Deployed DinoV3-based object detection at 65.3 mAP, serving inference at 10ms via **Python + FastAPI, MySQL**, and **AWS** (SageMaker, EC2, EKS) with CI/CD and **Docker**.

Computer Vision Engineer | Institute of Neurosciences, India | Onsite JAN 2022 – DEC 2023

- Designed and implemented custom 3D Computer Vision solutions for Neurological disorder analysis (**PyTorch, Python, OpenCV, Numpy**), improving diagnosis speed by 40% and reaching 88.68% accuracy on clinical scans.
- Created custom data processing and ingestion (ETL) pipelines for handling massive MRI, CT images reducing the model training data creation time by 80%.

Machine Learning Engineer | Mercedes-Benz Research & Development India, India | Onsite AUG 2018 – SEP 2019

- Developed and optimized a computer vision pipeline for Vulnerable Road User (VRU) detection by fine-tuning **YOLOv3** on large-scale annotated datasets, attaining detection mAP50 of 83% at real-time inference speeds.
- Designed & deployed model training, evaluation, and validation workflows in **AWS**.
- Built automated quality assurance (**QA**) and evaluation frameworks using Python, improving annotation throughput by 20% and data quality by 25%.

Systems Engineer | Tata Consultancy Services, India | Onsite AUG 2014 – APR 2015

- Maintained high-availability backend systems (Java, MySQL) for TCS's internal website, ensuring seamless functionality for over 100,000 users. Streamlined feature release cycle, reducing deployment time by 20%.

## EDUCATION

PhD | Jadavpur University, India FEB 2020 – APR 2025

M.Tech. Computer Science & Engineering | Jadavpur University, India | GPA - 8.83 AUG 2016 – JUN 2018

B.Tech. Computer Science & Engineering | Techno India, India | GPA - 8.81 AUG 2010 - MAY 2014

## SKILLS

|                         |                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Languages:              | Python, C++, SQL                                                                                                                    |
| LLM/Agent Systems:      | Multi Agent Orchestration (Crew AI, Google ADK, LangGraph), Agentic Memory, LangChain, LLM, APIs (OpenAI, Gemini, Claude), DeepEval |
| Frameworks / Libraries: | FastAPI, Flask, PyTorch, Scikit-learn, OpenCV, Open3D, PyTest,                                                                      |
| Cloud & Ops:            | Google Cloud Platform (GCP), Amazon Web Services (AWS), Git, Docker, MLFlow, Github Actions                                         |